The White Queen 600-cell

An exact atlas of the slices and shadows of the 600-cell

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The ideas are DICTO's. Computation and proofs with Claude (Anthropic).

Abstract. The 600-cell is the four-dimensional solid built from 600 regular tetrahedra, twenty round every vertex. DICTO asked for a definitive, proved picture of it. We give a complete exact atlas of its three-dimensional slices in its four main directions (through a vertex, the middle of an edge, the centre of a triangle and the centre of a cell). Every slice is computed exactly, in numbers of the form a + b√5 with a and b rational. Between two neighbouring vertex heights a slice keeps one pattern of faces, so the cuts at the vertex heights and one cut inside each gap give every slice pattern there is: 62 patterns on one side of the centre (9, 17, 21 and 15 in the four directions). Exactly twelve are made only of triangles, and we identify each one. The largest slices have 144 vertices. With the four shadows of the 600-cell, this completes the atlas. Each shadow resolves exactly into two halves, its front cells and its back cells, each filling it once; along the six turnings between the main views, the set of cells in front changes at only 14 moments. Round one vertex the 600 cells form 15 shells, each of one kind of piece, the innermost a core of 20 Cleo-Icos. The slices as they are, convex and exact, with faces of many sizes, are the subject of this paper. A short side note records what happens when a slice is forced to have equal edges: of six large all-triangle slices only one can be. That slice is Humpty Dumpty, a convex egg of 92 triangles; its forced equal-edged form, Fallen Humpty, has lost the egg's rounded shape.

1. Begin at the beginning

> *“Begin at the beginning,” the King said gravely, “and go on till you come to the end: then stop.” — Alice's Adventures in Wonderland, ch. 12*

That is the plan of this paper: start at the centre of the 600-cell and cut it, height by height, until the cut leaves it at the tip. DICTO's question was whether the 600-cell could be assembled into a definitive picture, every part proved. Our previous paper (#62, Shadows of the Fourth Dimension) asked which of its slices have only triangular faces, and which of those can be rebuilt with every edge equal; here we map every slice. The question is old. Aristotle wrote that among solids the pyramid fills space (*On the Heavens* III.8); his commentators argued over how many tetrahedra meet round a point, twelve or twenty, and Thomas Bradwardine saw that the icosahedron cuts from its centre into twenty equal tetrahedra without deciding whether they are regular (Lagarias and Zong, 2012; our #44, Aristotle's Pyramid). They are not: they are the Cleo-Icos. Kepler nested the Platonic solids in shells (*Mysterium Cosmographicum*, 1596). In the 600-cell, found by Schläfli about 1852, twenty regular tetrahedra do close round every vertex, and §6 shows its 600 cells in shells round such a core. The title is DICTO's: the 600-cell belongs to the White Queen of *Through the Looking-Glass*, who practised believing impossible things (§7).

2. Method

The 600-cell is taken with its 120 vertices at distance 2 from the centre: the 8 permutations of (±2, 0, 0, 0), the 16 points (±1, ±1, ±1, ±1) and the 96 even permutations of (±φ, ±1, ±1/φ, 0), where φ = (1+√5)/2. Its edges have length √5 − 1. All arithmetic is exact in $\mathbb{Q}(\sqrt{5})$. For each direction d, the height of a point x is x·d, and we sort the vertices by height. The slice at height h is the convex hull of the vertices at that height together with the points where edges of the 600-cell cross that height. Hulls are proposed in floating point, and every face is then verified exactly.

> **Lemma (one pattern per gap).** Let $h_1 < h_2$ be two neighbouring vertex heights. At every height strictly between them the slice has the same pattern of faces. *Proof.* An edge of the 600-cell crosses the cut at height h exactly when its two ends lie on opposite sides of h. No vertex has a height strictly between h_1 and h_2, so as h moves through the gap no vertex changes side, and the same edges cross the cut. Inside the gap each face of the slice is the cross-section of one tetrahedral cell, with its corners at the crossing points on that cell's edges, and two faces share an edge of the slice exactly when their cells share a triangle that is cut. Which cells and triangles are cut, and through which edges, is therefore the same throughout the gap, and so is the pattern of faces. Hence the cuts at the vertex heights and one cut inside each gap give every slice pattern. The lemma is about the pattern only: edge lengths and angles change with the height.

In each direction, d is the vertex, the edge midpoint, the triangle centre or the cell centre itself, not a unit vector, so heights compare within one direction only. The 600-cell is centrally symmetric, so the far half of each direction mirrors the near half; we list the near half, from the centre to the tip.

3. The atlas

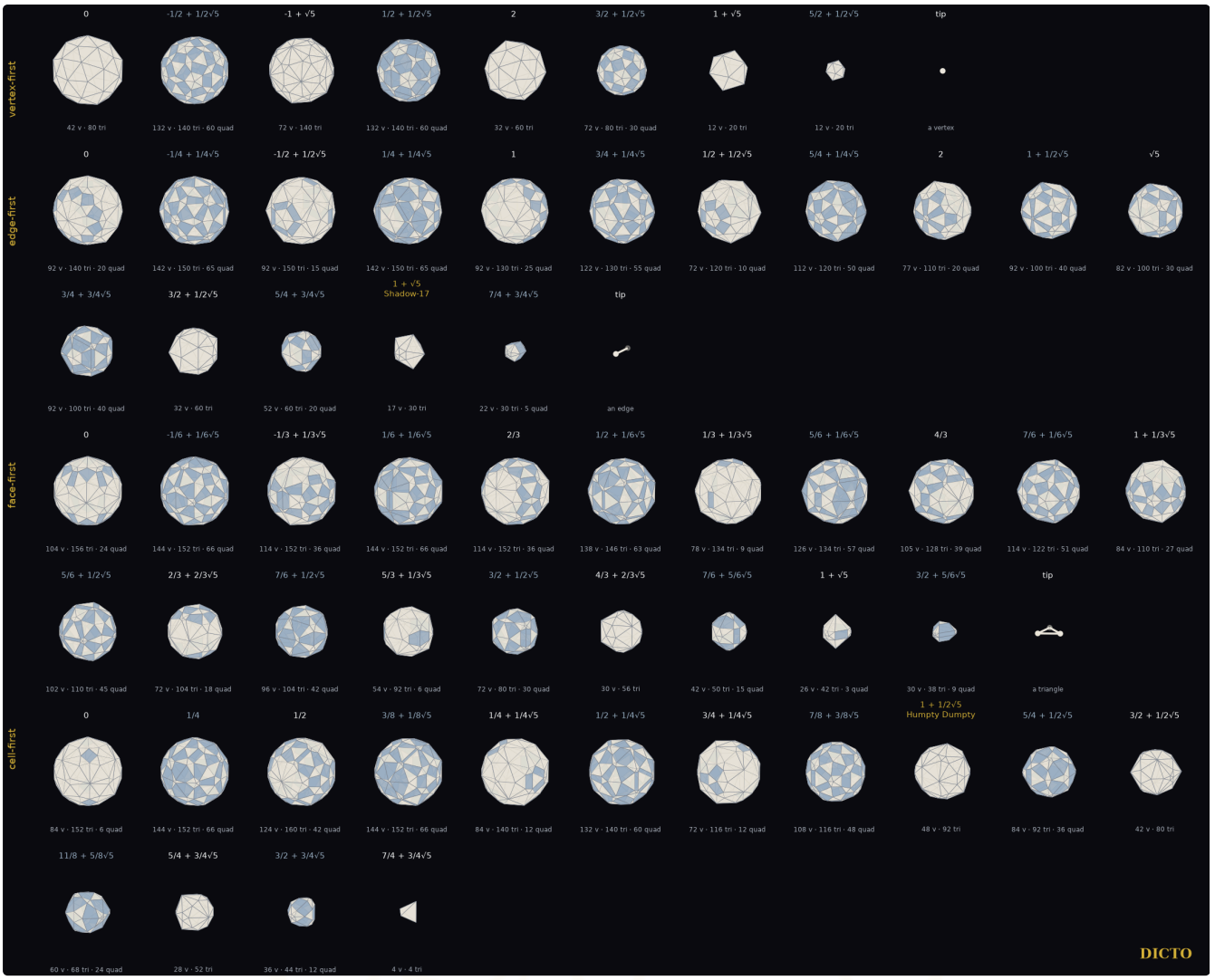


Figure 1. Every slice pattern of the 600-cell, from the centre to the tip, read left to right and row by row, in the four directions; one scale throughout. White labels: slices at a vertex height; blue labels: the pattern that holds over the whole gap to the next height. Triangles ivory, four-sided faces blue. Gold: Shadow-17 (#62) and Humpty Dumpty (§7).

Direction	Vertex heights (centre to tip)	Gaps	Slice patterns	Tip	Largest slice
vertex-first	5	4	9	a vertex	132 vertices
edge-first	9	8	17	an edge	142 vertices
face-first	11	10	21	a triangle	144 vertices
cell-first	8	7	15	a tetrahedron	144 vertices
total			**62**		

Every slice has only triangles and four-sided faces. Each face is the cross-section of one tetrahedral cell, a triangle or a four-sided face, or else a triangle of the 600-cell lying in the cut. A four-sided face appears exactly where the cut crosses a cell through four of its edges. The atlas, which cuts each gap at its middle height, has 1636 four-sided faces, and exactly 46 of them are squares, checked exactly. The squares are exactly the cuts through the middle of a cell, parallel to two opposite edges: each is the equator of the regular octahedron that joins the midpoints of that cell's six edges (edge $(\sqrt{5}-1)/2 = 1/\varphi$). Moved off the middle of the cell, such a cut is a rectangle. The 104 kites are explained in the same way: a kite appears exactly

where a cell sits symmetrically across the cut, turned so that its four corners lie at four different heights, −p, −q, q and p from the cut. The half-turn of the cell that exchanges the corners at −p and p, and those at −q and q, reverses the heights and so maps the cut onto itself; on the four-sided face it fixes two opposite corners and exchanges the other two, a mirror along a diagonal (checked exactly: 104 kites, 104 such cells). The remaining faces are rectangles, isosceles trapezia and general quadrilaterals.

4. The twelve all-triangle slices

> **Theorem.** Exactly twelve of the 62 slice patterns are made only of triangles: eleven at vertex heights and one inside a gap.

Direction	Height	Vertices, triangles	What it is
vertex-first	0 (centre)	42, 80	the central slice; equal-edged form: the 12-dented icosahedron (#62)
vertex-first	√5 − 1	72, 140	no equal-edged form with its full symmetry (#62)
vertex-first	2	32, 60	equal-edged forms: excavated and augmented dodecahedron (#62)
vertex-first	1 + √5	12, 20	a regular icosahedron: the twelve neighbours of the tip vertex
vertex-first	the gap from 1 + √5 to 4	12, 20	a regular icosahedron: the vertex figure, cut across the twelve edges that meet the tip
edge-first	(3+√5)/2	32, 60	no equal-edged form with its full symmetry (#62)
edge-first	1 + √5	17, 30	Shadow-17: two J11 glued on their pentagons (#62)
face-first	(4+2√5)/3	30, 56	no equal-edged form with its full symmetry (#62)
cell-first	1 + √5/2	48, 92	**Humpty Dumpty**, a convex egg; forced to equal edges it becomes Fallen Humpty (§7)
cell-first	(3+√5)/2	42, 80	no equal-edged form with its full symmetry (#62)
cell-first	(5+3√5)/4	28, 52	no equal-edged form with its full symmetry (#62)
cell-first	(7+3√5)/4 (tip)	4, 4	a regular tetrahedron: the cell at the tip

The other 50 slice patterns all contain four-sided faces. Of the twelve, the two regular icosahedra and the tetrahedron are the 600-cell's own vertex figure and cell; the other nine are the all-triangle patterns studied in #62.

5. The four shadows

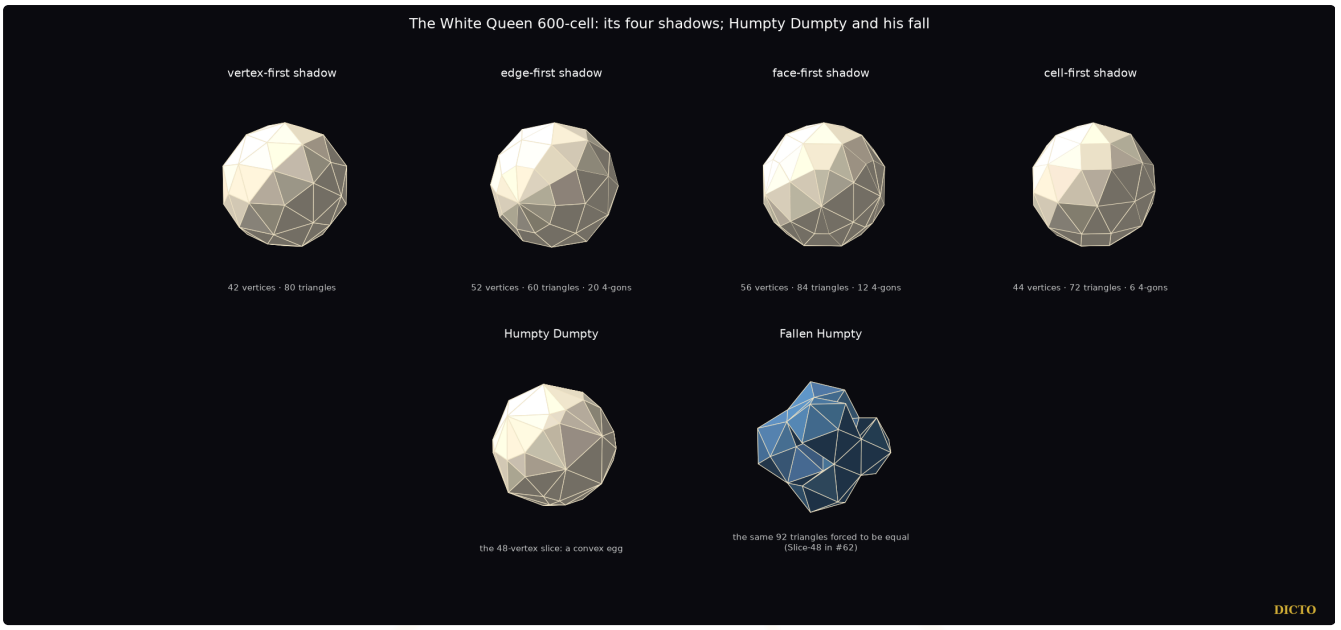


Figure 2. Top: the four shadows of the 600-cell. Bottom: Humpty Dumpty, the 48-vertex cell-first slice, a convex egg with edges of several lengths; and Fallen Humpty, the same 92 triangles forced to be equal (§7).

The shadows are the projections of the 600-cell along the four directions, and all four are convex. The vertex-first shadow has 42 vertices and 80 triangles, the same pattern as the central slice (#62). The edge-first shadow has 52 vertices, 60 triangles and 20 four-sided faces; the face-first shadow 56 vertices, 84 triangles and 12 four-sided faces; the cell-first shadow 44 vertices, 72 triangles and 6 four-sided faces.

6. The shadows resolved, and the Cleo shells

Halves. In each main view, every cell of the 600-cell faces the viewer, faces away, or is seen side-on, decided exactly by the sign of (cell centre)·d. The front cells fill the shadow exactly once, without gaps or overlaps, and so do the back cells. Checked exactly: in each view the projected volumes of the front cells add up to the volume of the shadow, and so do those of the back cells. The side-on cells flatten into the shadow's surface. The front cells together are the shadow's front layer. Two front cells are of one kind when their projections have the same six edge lengths.

View	Front	Back	Side-on	Kinds of front piece
vertex-first	270	270	60	7
edge-first	290	290	20	22
face-first	288	288	24	33
cell-first	273	273	54	20

The six turnings. Any two of the four main views span a plane in which the view can turn from one to the other: six turnings in all. Along a turning the front layer changes only at a moment when a cell turns side-on. At such a moment cells change sides in opposite pairs, a cell coming forward as the cell opposite it goes back, so the number in front stays the same. Between two such moments the front layer keeps one pattern (the shadow's counterpart of the lemma of §2). Computed exactly in $\mathbb{Q}(\sqrt{5})$, the six turnings (vertex–edge 18°, vertex–triangle 20.905°, vertex–cell 22.239°, edge–triangle 10.812°, edge–cell 13.283°, triangle–cell

7.761°) contain 14 such moments and 20 patterns between them; with the four main views, this accounts for every view along the turnings. Away from the moments the front layer holds 300 cells, or 298 or 294 where 4 or 12 cells stay side-on along the whole turning. One moment on the vertex–cell turning lies 15.52° from the vertex-first view, at the angle whose cosine is (1 + 3√5)/8: that is 180° minus the 600-cell's dihedral angle, 164.48°.

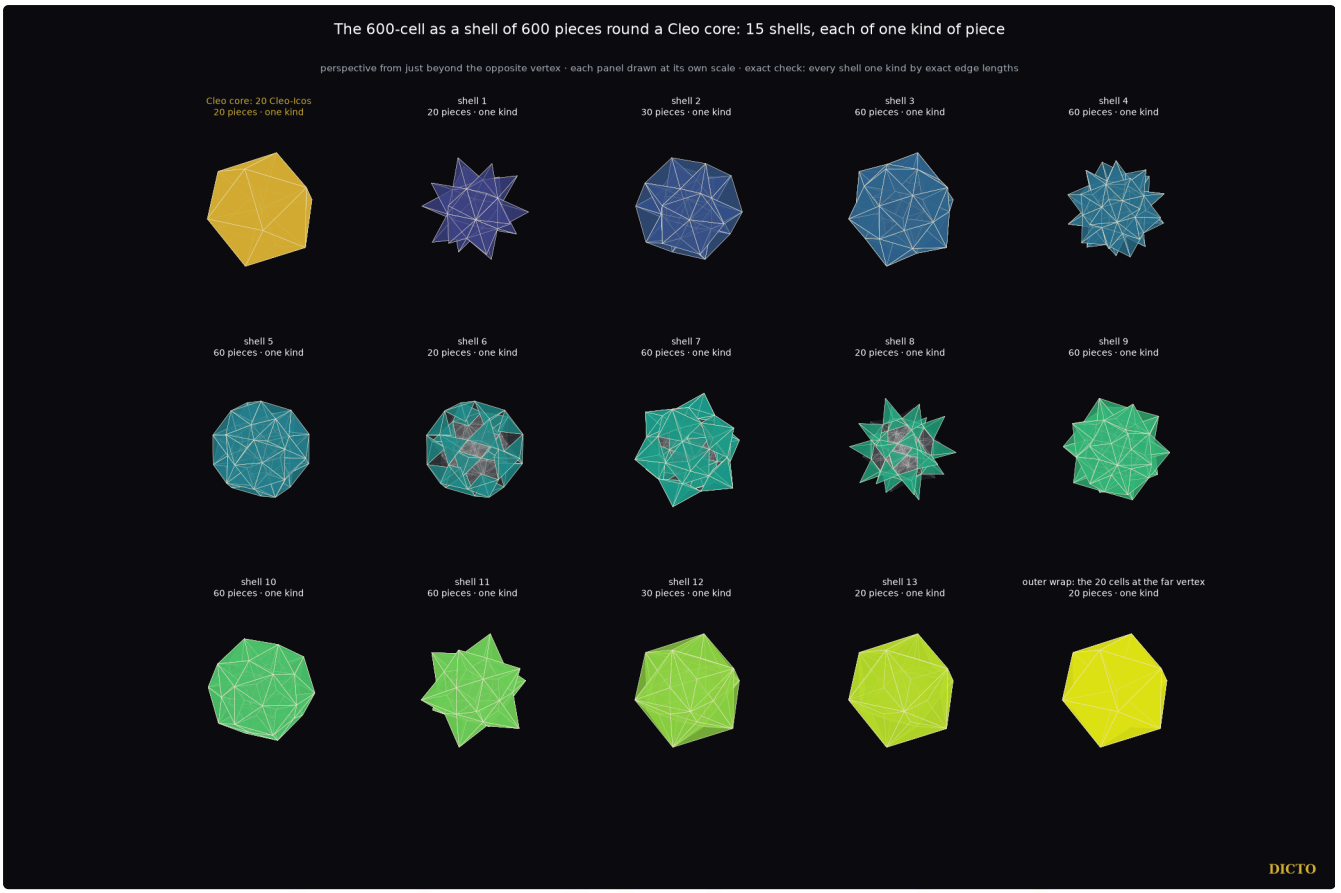


Figure 3. The 600-cell's 600 cells in 15 shells round one vertex, seen in perspective from just beyond the opposite vertex; each panel shows one shell solid and the shells inside it faint, at its own scale. Gold: the core of 20 Cleo-Icos.

The Cleo shells. Grouped by the height of their centres along one vertex, the 600 cells form 15 shells of 20, 20, 30, 60, 60, 60, 20, 60, 20, 60, 60, 60, 30, 20 and 20 cells. DICTO noticed that this list reads the same backwards. The angles from the vertex to the shells' cell centres mirror it too: 22.24°, 37.76°, 45°, 55.11°, 69.30°, 77.38°, 82.24°, 90°, then 97.76° and on to 157.76°, and each shell's angle and its mirror shell's angle add up to exactly 180°. Both follow from the 600-cell's symmetry through its centre, which carries shell k to shell 14 − k and the vertex to the opposite vertex. Seen in perspective from just beyond the opposite vertex (eye at height −21/10 when the vertex is at 2), the cells of each shell are all of one kind: they have the same six edge lengths, checked exactly. The core is the innermost shell, the 20 cells round the vertex. In this view their common corner is the centre, and their twelve other corners, the vertex's twelve neighbours, form a regular icosahedron about it (§4); so each core cell is the central pyramid of a regular icosahedron, the piece DICTO named the Cleo-Icos (#19). Mosseri and Sadoc flatten the 600-cell into ordinary space to describe icosahedral order in materials (§8); the icosahedron of 20 Cleo-Icos is the 600-cell's vertex cluster brought flat. In DICTO's words, the Cleo-Icos is the 600-cell's original tetrahedron as it must look in flat space: twenty of them fill the regular icosahedron exactly. The shells also explain the vertex-

first shadow: its front layer is exactly the core and the next six shells (270 cells), its side-on cells are the middle shell (60), and its back layer is the other seven shells.

DICTO's radial cell projection. In the Polyhedraverse app, DICTO's radial cell projection (RCP) builds the 600-cell cell by cell, each new cell folded across a face of one already placed. It is his rediscovery of the classical reflection construction (Coxeter). The fold between neighbouring cells is the angle with cosine (1 + 3√5)/8, 15.52°: the same angle as the moment on the vertex–cell turning above. Counted in steps from one cell to a face-to-face neighbour, the 600 cells fall into 16 shells of 1, 4, 12, 24, 36, 52, 68, 76, 78, 72, 64, 56, 40, 12, 4 and 1 cells, checked here on the exact model (schlegel_shells.py). The 600-cell thus unfolds in three ways from one exact model: in DICTO's 16 steps from a cell, in 31 layers of height from a cell, and in the 15 Cleo shells round a vertex.

7. A side note: six impossible things, and Humpty Dumpty's fall

> *"Why, sometimes I've believed as many as six impossible things before breakfast." — the White Queen, Through the Looking-Glass, ch. 5*

Six of the all-triangle slices are large, with 28 to 72 vertices, and match none of the eight convex deltahedra: the slices with 72 (vertex-first), 32 (edge-first), 30 (face-first), and 48, 42 and 28 (cell-first) vertices. #62 asked of each whether its triangles can all be made equal while its full symmetry is kept. For five of them this is impossible. For the 48-vertex slice it is possible in exactly one way, and the result has a name.

> *"All the King's horses and all the King's men / Couldn't put Humpty Dumpty in his place again." — Through the Looking-Glass, ch. 6*

The real 48-vertex slice, at height 1 + √5/2 in the cell-first direction, is **Humpty Dumpty** (DICTO's name): a convex, egg-like solid of 92 triangles, with edge lengths in ratios up to (5+√5)/2 ≈ 3.62 (Figure 2). Forced to have all 138 edges equal, it becomes **Fallen Humpty** (placeholder name; Slice-48 in #62): a non-convex solid that folds in and out, with dihedral angles from 81.75° to 243.20°, where the egg was smooth. As DICTO remarked, the rhyme describes the situation exactly: the equal triangles put him together again, but not in his place. What the 600-cell shows is the egg; Fallen Humpty is what our demand for equal triangles makes of it.

> *"When I use a word," Humpty Dumpty said in rather a scornful tone, "it means just what I choose it to mean—neither more nor less." — ch. 6*

8. Prior art and credit

Alicia Boole Stott (1900; encouraged by P. H. Schoute) studied series of three-dimensional sections of the regular four-dimensional polytopes and made complete sets of models of the sections of the 600-cell and the 120-cell; two sets of 600-cell section models are kept at the University of Groningen, and a further set is at the Faulkes Institute, Cambridge. Coxeter's *Regular Polytopes* treats the polytopes and the vertex-first sections. The directions and number of sections in Boole Stott's series are not stated in the sources we checked, and we did not read her paper; her work is the precedent for this atlas. Mosseri and Sadoc flatten the 600-cell into ordinary space to describe icosahedral order in materials (C. R. Physique 15, 2014, doi:10.1016/j.crhy.2013.09.006); that its cells then become Cleo-Icos was noted in the Cleo-Icos reading study of 14 October 2026, made on DICTO's question. Counts of cells by

layer round a vertex of the 600-cell are a natural tabulation and may be known; we did not find them in the sources checked. We did not find in the sources checked: an exact atlas in all four directions including the gaps between vertex heights, the list of all-triangle slices, or Humpty Dumpty.

> **Sources checked** (27 October 2026). *History*: J. C. Lagarias and C. Zong, "Mysteries in packing regular tetrahedra", Notices of the AMS 59 (2012), as read for #44 (21 October 2026), for Aristotle, his commentators and Bradwardine; Kepler, *Mysterium Cosmographicum* (1596), and Schläfli's discovery of the 600-cell are cited for their well-known contents, not re-read. *Web*: Wikipedia, "Alicia Boole Stott"; G. W. Hart, "Models of sections of the 600-cell, made by Alicia Boole Stott" (math.stonybrook.edu); search for the 1900 paper (Proofwiki, MacTutor); Wikipedia, "600-cell"; Wolfram Demonstrations, "Sections of the 16-Cell"; arXiv 1912.06156, "The Geometry of H4 Polytopes" (title only); KNAW digital library (dwc.knaw.nl, title page only). *Library*: the Dodgson texts (Gutenberg) for every quotation, checked word for word. Coxeter, *Regular Polytopes* (3rd ed. 1973), cited for its well-known contents, not re-read.

9. Conclusion

The 600-cell has 62 slice patterns between its centre and its tips in its four main directions (the far half mirrors the near half), all computed exactly here; exactly twelve are made only of triangles, and each is identified. With the four shadows this is a complete, proved atlas of the White Queen's solid, every slice shown as it really is. Its shadows resolve into exact halves, and its 600 cells stand in 15 shells round one vertex, each shell of one kind of piece, the innermost a core of 20 Cleo-Icos. Its most graceful slice is Humpty Dumpty, the 48-vertex egg. Forcing equal edges, the question of #62, succeeds for that one large slice only, and its result, Fallen Humpty, is less graceful than the egg it came from.

> **Version 2** (28 October 2026): §6 adds DICTO's observation that the fifteen Cleo shells read the same backwards, with mirrored angles adding up to 180°.

> **Data and code.** Study 2026-10-27-white-queen-600 (atlas600.py, render_atlas.py; data/atlas600.json, atlas600.pkl), using the exact modules of study 2026-10-27-4d-shadows (q5.py, polytopes.py, shadows.py) and its Slice-48 data (form_V48_full_36.json). References: DICTO, Shadows of the Fourth Dimension (#62), DICTO DOI 10.5281/zenodo.23687223; DICTO, AXE, CLEO and FUJI: the Gap Pieces of the Icosahedron (#19), DICTO DOI 10.5281/zenodo.23351291. Further study files: squares_midcuts.py, front_cells_exact.py, shadow_turnings_exact.py, cleo_shells.py (data/front_cells_exact.json, shadow_turnings_exact.json, cleo_shells.json). Quotations: L. Carroll, *Alice's Adventures in Wonderland* (1865), *Through the Looking-Glass* (1871), Project Gutenberg texts.